**CAMBRIDGE INTERNATIONAL EXAMINATIONS**

**GCE Ordinary Level**

# MARK SCHEME for the May/June 2013 series

## 5054 PHYSICS

**5054/41** Paper 4 (Alternative to Practical), maximum raw mark 30

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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| | | | | | |
|---|---|---|---|---|---|
| 1 | (a) | (i) | 1.046 or 1.05 seen | C1 | |
| | | | 1.05 s cao unit required | A1 | [2] |
| | | (ii) | large difference in raw data/<br>reaction time has large variation | B1 | [1] |
| | | (iii) | one drops, one times | B1 | |
| | | | synchronise/countdown or signal explained | B1 | [2] |
| | (b) | (i) | axes: correct way round, labelled quantity and unit | B1 | |
| | | | scales: more than ½ grid, linear, not awkward<br>e.g. *y*-axis: 2 cm ≡ 0.1 s *x*-axis: 2 cm ≡ 2 | B1 | |
| | | | points plotted accurately within ½ small square<br>neat crosses or small points (in circle) | B1 | |
| | | | smooth curve of best fit drawn | B1 | [4] |
| | | (ii) | $x \times y$ seen with substitution of one set of values from table or graph | B1 | |
| | | | two values calculated and not equal comment | B1 | |
| | | | 'two values of *t* same in table so $x \times y$ not constant' scores 2/2 | | [2] |
| | (c) | | sensible suggestion, e.g.<br>holds arm horizontal<br>stands in same place and uses a marker<br>in front of mirror and uses image | B1 | [1] |
| | (d) | | mass or weight cao | B1 | [1] |
| | (e) | | changes surface area/air resistance<br>cases will not stack | B1 | [1] |
| | (f) | | heavier so air resistance has little/same effect<br>smaller % change in mass<br>due to uncertainty in timing/height | B1 | [1] |

**[Total: 15]**

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**2** **(a)** **(i)** **1.** three resistors drawn in series B1 [1]

**2.** 470 Ω cao unit required B1 [1]

**(ii)** three resistors drawn in parallel B1 [1]

**(b)** 180 Ω cao unit required B1 [1]

**[Total: 4]**

**3** **(a)** 22(.0) °C unit required B1 [1]

**(b)** **(i)** all the oil is heated/
all oil below water surface/
uniform heating of oil B1 [1]

**(ii)** temperature rises B1
then falls B1 [2]

**(iii)** avoid parallax error/good explanation
reads top of meniscus
aligns scale with liquid column B1 [1]

**(c)** smooth concave curve B1
asymptotes to above zero B1 [2]

**[Total: 7]**

**4** freely suspends lamina from hole B1
correct use of plumb-line shown on diagram B1
line marked on lamina B1
repeated from different hole and find where lines cross B1 [4]

alternative experiments:
balancing on a ruler can score points 1, 3 and 4 (max. 3)
finding balance point by trial and error on a pin (max. 1)

**[Total: 4]**

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